

Gaussian Measures from Positive Semi-Definite Kernels: Formal Definitions and the RKHS Prior Correspondence

Abstract

We provide formal definitions for constructing Gaussian measures from positive semi-definite kernels and establish the precise correspondence between these measures and the informal prior “ $p(f) \propto \exp(-\frac{1}{2}\|f\|_{\mathcal{H}_K}^2)$ ” via the Onsager-Machlup functional.

Contents

1 Setup and Notation

Let $(\mathcal{X}, \mathcal{A}, P_X)$ be a probability space where $\mathcal{X}$ is the input domain. Let $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a symmetric positive semi-definite (PSD) kernel satisfying the integrability condition:

$$\int_{\mathcal{X}} K(x, x) dP_X(x) < \infty. \quad (1)$$

We work primarily in the Hilbert space $L^2(P_X)$ with inner product

$$\langle f, g \rangle_{L^2} = \int_{\mathcal{X}} f(x)g(x) dP_X(x). \quad (2)$$

2 The Integral Operator

Definition 2.1 (Integral Operator). The integral operator $T_K : L^2(P_X) \rightarrow L^2(P_X)$ associated with kernel K is defined by

$$(T_K g)(x) = \int_{\mathcal{X}} K(x, y) g(y) dP_X(y). \quad (3)$$

Theorem 2.2 (Spectral Decomposition). *The operator T_K is self-adjoint, positive, and trace-class. It admits a spectral decomposition*

$$T_K \phi_k = \mu_k \phi_k, \quad k \in \mathbb{N}, \quad (4)$$

where:

- (i) $\mu_1 \geq \mu_2 \geq \dots > 0$ are the positive eigenvalues (counted with multiplicity),
- (ii) $\{\phi_k\}_{k=1}^{\infty}$ is an orthonormal basis of $\overline{\text{Range}(T_K)} \subseteq L^2(P_X)$,
- (iii) $\sum_{k=1}^{\infty} \mu_k = \text{Tr}(T_K) = \int_{\mathcal{X}} K(x, x) dP_X(x) < \infty$.

The kernel admits the Mercer representation

$$K(x, y) = \sum_{k=1}^{\infty} \mu_k \phi_k(x) \phi_k(y), \quad (5)$$

with convergence in $L^2(P_X \otimes P_X)$.

3 The Reproducing Kernel Hilbert Space

Definition 3.1 (RKHS via Spectral Representation). The reproducing kernel Hilbert space (RKHS) $\mathcal{H}_K$ associated with K is

$$\mathcal{H}_K = \left\{ f = \sum_{k=1}^{\infty} \theta_k \phi_k : \|f\|_{\mathcal{H}_K}^2 := \sum_{k=1}^{\infty} \frac{\theta_k^2}{\mu_k} < \infty \right\}, \quad (6)$$

equipped with the inner product

$$\langle f, g \rangle_{\mathcal{H}_K} = \sum_{k=1}^{\infty} \frac{\theta_k \eta_k}{\mu_k}, \quad \text{where } f = \sum_k \theta_k \phi_k, \quad g = \sum_k \eta_k \phi_k. \quad (7)$$

Proposition 3.2 (Operator Characterization). The RKHS admits the equivalent characterization

$$\mathcal{H}_K = \text{Range}(T_K^{1/2}) \quad (8)$$

with norm

$$\|f\|_{\mathcal{H}_K} = \|T_K^{-1/2} f\|_{L^2}. \quad (9)$$

The inclusion $\mathcal{H}_K \hookrightarrow L^2(P_X)$ is continuous and dense in $\overline{\text{Range}(T_K)}$.

Remark 3.3. The space $\mathcal{H}_K$ is “smaller” than $L^2(P_X)$: for $f \in \mathcal{H}_K$, the coefficients $\theta_k = \langle f, \phi_k \rangle_{L^2}$ must decay faster than $\sqrt{\mu_k}$.

4 The Gaussian Measure

Definition 4.1 (Gaussian Measure via Characteristic Functional). The *Gaussian measure* μ_K on $L^2(P_X)$ is the unique Borel probability measure with characteristic functional

$$\hat{\mu}_K(g) := \int_{L^2(P_X)} e^{i\langle g, f \rangle_{L^2}} d\mu_K(f) = \exp\left(-\frac{1}{2} \langle g, T_K g \rangle_{L^2}\right) \quad (10)$$

for all $g \in L^2(P_X)$.

Theorem 4.2 (Existence and Characterization). The measure μ_K exists, is unique, and satisfies the following:

(i) **Karhunen-Loève representation:**

$$f \sim \mu_K \iff f = \sum_{k=1}^{\infty} \sqrt{\mu_k} \xi_k \phi_k, \quad \xi_k \stackrel{\text{iid}}{\sim} N(0, 1) \quad (11)$$

where the series converges in $L^2(\Omega; L^2(P_X))$.

(ii) **Mean and covariance:**

$$\mathbb{E}_{\mu_K}[f] = 0, \quad (12)$$

$$\mathbb{E}_{\mu_K}[\langle f, g \rangle_{L^2} \langle f, h \rangle_{L^2}] = \langle g, T_K h \rangle_{L^2}, \quad \forall g, h \in L^2(P_X). \quad (13)$$

Equivalently, $\mathbb{E}_{\mu_K}[f \otimes f] = T_K$ as operators.

(iii) **Finite-dimensional distributions:** For any $x_1, \dots, x_n \in \mathcal{X}$,

$$(f(x_1), \dots, f(x_n))^\top \sim N(0, K_{\mathbf{XX}}), \quad (14)$$

where $(K_{\mathbf{XX}})_{ij} = K(x_i, x_j)$.

Remark 4.3 (Gaussian Process Equivalence). The measure μ_K is the law of the Gaussian process $f \sim \mathcal{GP}(0, K)$.

5 The Cameron-Martin Theorem

The following theorem establishes $\mathcal{H}_K$ as the Cameron-Martin space of μ_K , providing the fundamental link between the RKHS and the Gaussian measure.

Theorem 5.1 (Cameron-Martin). *Let μ_K be the Gaussian measure on $L^2(P_X)$ and $\mathcal{H}_K$ its associated RKHS. Then:*

(i) **Null RKHS:** Sample paths lie outside the RKHS almost surely:

$$\mu_K(\mathcal{H}_K) = 0. \quad (15)$$

(ii) **Admissible shifts:** For $h \in L^2(P_X)$, define the translated measure $\mu_K^h(A) := \mu_K(A - h)$. Then

$$\mu_K^h \ll \mu_K \iff h \in \mathcal{H}_K. \quad (16)$$

That is, $\mathcal{H}_K$ is exactly the set of admissible shifts preserving absolute continuity.

(iii) **Radon-Nikodym derivative:** For $h \in \mathcal{H}_K$,

$$\boxed{\frac{d\mu_K^h}{d\mu_K}(f) = \exp\left(\langle h, f \rangle_{\mathcal{H}_K} - \frac{1}{2}\|h\|_{\mathcal{H}_K}^2\right)} \quad (17)$$

where the pairing $\langle h, f \rangle_{\mathcal{H}_K}$ extends to $f \in L^2(P_X)$ via

$$\langle h, f \rangle_{\mathcal{H}_K} = \sum_{k=1}^{\infty} \frac{\eta_k \theta_k}{\mu_k}, \quad h = \sum_k \eta_k \phi_k, \quad f = \sum_k \theta_k \phi_k. \quad (18)$$

Corollary 5.2. *The RKHS norm $\|\cdot\|_{\mathcal{H}_K}$ measures the “cost” of deterministic shifts under the Gaussian measure.*

6 The Onsager-Machlup Functional

The formal expression “ $p(f) \propto \exp(-\frac{1}{2}\|f\|_{\mathcal{H}_K}^2)$ ” is made rigorous by the following theorem.

Theorem 6.1 (Onsager-Machlup). *Let μ_K be a Gaussian measure on a Banach space $\mathcal{B} \supseteq \mathcal{H}_K$ with Cameron-Martin space $\mathcal{H}_K$. Let $\|\cdot\|_{\mathcal{B}}$ be a measurable norm satisfying appropriate regularity conditions.¹ Then for any $f, g \in \mathcal{H}_K$:*

$$\lim_{\epsilon \rightarrow 0} \frac{\mu_K(\{h \in \mathcal{B} : \|h - f\|_{\mathcal{B}} < \epsilon\})}{\mu_K(\{h \in \mathcal{B} : \|h - g\|_{\mathcal{B}} < \epsilon\})} = \exp\left(-\frac{1}{2}\|f\|_{\mathcal{H}_K}^2 + \frac{1}{2}\|g\|_{\mathcal{H}_K}^2\right) \quad (19)$$

Corollary 6.2 (RKHS Norm as Log-Density). *Taking $g = 0$:*

$$\lim_{\epsilon \rightarrow 0} \frac{\mu_K(B_\epsilon(f))}{\mu_K(B_\epsilon(0))} = \exp\left(-\frac{1}{2}\|f\|_{\mathcal{H}_K}^2\right). \quad (20)$$

This is the precise sense in which

$$“p(f) \propto \exp\left(-\frac{1}{2}\|f\|_{\mathcal{H}_K}^2\right).” \quad (21)$$

Remark 6.3. The function $f \mapsto \frac{1}{2}\|f\|_{\mathcal{H}_K}^2$ is called the *Onsager-Machlup functional* of μ_K . It plays the role of a “negative log-density” for the Gaussian measure.

7 Connection to Bayesian Inference and Kernel Ridge Regression

Theorem 7.1 (Posterior and MAP Estimator). *Suppose we observe*

$$Y_i = f(X_i) + \varepsilon_i, \quad i = 1, \dots, n, \quad (22)$$

where $\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$, $X_i \stackrel{\text{iid}}{\sim} P_X$, and the prior is $f \sim \mu_K$. Then:

- (i) *The posterior $\mu_K^{(n)}(\cdot \mid \mathbf{Y}, \mathbf{X})$ is a Gaussian measure.*
- (ii) *The posterior mean coincides with the kernel ridge regression estimator:*

$$\hat{f} = \arg \min_{f \in \mathcal{H}_K} \left\{ \frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - f(X_i))^2 + \|f\|_{\mathcal{H}_K}^2 \right\} \quad (23)$$

(iii) *In closed form:*

$$\hat{f}(\cdot) = \mathbf{k}(\cdot)^\top (K_{\mathbf{X}\mathbf{X}} + \sigma^2 I)^{-1} \mathbf{Y}, \quad (24)$$

where $\mathbf{k}(x) = (K(x, X_1), \dots, K(x, X_n))^\top$.

Remark 7.2. The RKHS norm appears as the regularizer in the MAP estimator precisely because it is the Onsager-Machlup functional of the GP prior. This establishes the equivalence:

$$\text{GP regression with covariance } K \longleftrightarrow \text{Kernel ridge regression with RKHS } \mathcal{H}_K.$$

¹Specifically, $\|\cdot\|_{\mathcal{B}}$ should be measurable and the small ball probabilities $\mu_K(B_\epsilon(0))$ should satisfy certain asymptotic conditions; see Zeitouni (1989).

8 Sum Kernels

For $K = \sum_{j=1}^J a_j K_{A_j}$ with $a_j > 0$ and each K_{A_j} PSD:

Theorem 8.1 (Convolution Structure). *The Gaussian measure satisfies*

$$\mu_K = \mu_{a_1 K_{A_1}} * \mu_{a_2 K_{A_2}} * \cdots * \mu_{a_J K_{A_J}}, \quad (25)$$

where $*$ denotes convolution of measures. Equivalently,

$$f \sim \mu_K \iff f = \sum_{j=1}^J f_j, \quad f_j \stackrel{\text{ind}}{\sim} \mu_{a_j K_{A_j}} \quad (26)$$

Theorem 8.2 (RKHS of Sum Kernel). *The RKHS of $K = \sum_{j=1}^J a_j K_{A_j}$ is*

$$\mathcal{H}_K = \mathcal{H}_{K_{A_1}} + \cdots + \mathcal{H}_{K_{A_J}} = \left\{ \sum_{j=1}^J f_j : f_j \in \mathcal{H}_{K_{A_j}} \right\} \quad (27)$$

with the infimal convolution norm:

$$\|f\|_{\mathcal{H}_K}^2 = \inf_{\substack{f = \sum_{j=1}^J f_j \\ f_j \in \mathcal{H}_{K_{A_j}}}} \sum_{j=1}^J \frac{1}{a_j} \|f_j\|_{\mathcal{H}_{K_{A_j}}}^2 \quad (28)$$

9 Summary: The Complete Correspondence

Given: PSD kernel $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$

$\Downarrow$

Integral operator: $T_K : L^2(P_X) \rightarrow L^2(P_X), \quad T_K \phi_k = \mu_k \phi_k$

$\Downarrow$

RKHS: $\mathcal{H}_K = \left\{ f = \sum_k \theta_k \phi_k : \sum_k \frac{\theta_k^2}{\mu_k} < \infty \right\}$

$\Downarrow$

Gaussian measure: $\hat{\mu}_K(g) = \exp \left(-\frac{1}{2} \langle g, T_K g \rangle_{L^2} \right)$

$\Downarrow$

Karhunen-Loève: $f \sim \mu_K \iff f = \sum_{k=1}^{\infty} \sqrt{\mu_k} \xi_k \phi_k, \quad \xi_k \stackrel{\text{iid}}{\sim} N(0, 1)$

$\Downarrow$

Onsager-Machlup: $\frac{\mu_K(B_\epsilon(f))}{\mu_K(B_\epsilon(0))} \xrightarrow{\epsilon \rightarrow 0} \exp \left(-\frac{1}{2} \|f\|_{\mathcal{H}_K}^2 \right)$

$\Downarrow$

Prior interpretation: $p(f) \propto \exp \left(-\frac{1}{2} \|f\|_{\mathcal{H}_K}^2 \right)$

10 References

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